Modular Carbon-Nanotube Tether Architecture for Space-Elevator Deployment: A Coupled Systems Analysis

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Abstract

Every published space-elevator architecture presumes a defect-free monolithic carbon-nanotube ribbon spanning 100000 km. This paper presents a modular alternative—variable-length CNT segments joined by nanobonded sleeve couplers—via a coupled analysis integrating taper geometry, joint reliability, repair logistics, dynamic stability, and lifecycle economics.

We resolve a taper-ratio discrepancy in the literature: all published values ($T_r \approx 1.6$ to $T_r \approx 37$) are recoverable from a single integral once the stress basis, material density, and base-area sizing are specified. Monte Carlo simulation (12,600 combinations $\times$ 10^5 trajectories) shows the modular architecture achieves >99.5% ten-year survival probability across the well-designed regime ($\eta_j \geq 0.88$, $p_{\text{det}} \geq 0.90$) *conditional on the assumed activation energy* $Q \approx 1.1$ eV. A Q-sensitivity sweep reveals that the hazard rate spans 10 orders of magnitude across $Q \in \{0.8, 1.1, 1.4\}$ eV: at $Q = 0.8$ eV all configurations fail catastrophically ($P_{\text{sys}} = 0$), while at $Q = 1.4$ eV no failures occur. Experimental characterization of Q for CNT sleeve bonds is therefore the single most important priority for validating these predictions.

A 2D Coriolis-coupled finite element model reveals a transverse fundamental period of 34.8 h and a multi-climber resonance separation rule at ∼35 h. The Edwards & Westling 72 h repair target is structurally constrained by inspection cadence, not depot coverage. Lifecycle NPV shows modular consistently outperforms monolithic, driven by a phased-construction revenue advantage contingent on a deployment strategy enabling partial operation at ∼60% completion.

The modular tether reframes the space-elevator challenge from materials perfection to systems engineering and maintainability.

1 Introduction

Earth-to-orbit transportation costs \$2,000–\$4,000/kg via chemical rockets, with approximately 400 t CO_2-equivalent per launch. A space elevator—a continuous tether from an equatorial anchor through geostationary orbit (GEO, 35786 km) to a counterweight beyond 100000 km—could reduce marginal launch costs by 1–2 orders of magnitude using electrically powered climbers with zero propellant.

The linchpin is the tether. Analytical tapering requires specific tensile strengths exceeding 30 $\text{GPa} \cdot \text{cm}^3/\text{g}$ over 100000 km. Carbon-nanotube yarns now achieve 80 GPa on centimeter gauges [2] and continuous kilometer-scale fibers at 4.1 $\text{N} \cdot \text{tex}^{-1}$ tensile strength (≈5.3 GPa at

$\rho = 1300$ kg/m^3; Niu et al. 12), narrowing the gap but leaving a significant shortfall relative to the assumed $\sigma_u = 50$ GPa baseline (see §2.2).

1.1 The Core Problem: Architecture, Not Materials

The core problem is not materials—it is architecture. Every published elevator design assumes a defect-free monolithic CNT ribbon. This assumption:

1. Demands planetary-scale manufacturing perfection with zero defect propagation over 100000 km;
2. Offers no mechanism to replace sections damaged by micrometeoroids, debris, or radiation;
3. Prevents incremental deployment, in-service upgrades, or phased construction;
4. Makes the entire system a single point of failure.

No published study has quantified joint reliability and load-path continuity in a segmented CNT tether at system level.

1.2 Research Objectives

O1. Design a modular CNT tether whose variable-length segments satisfy static and dynamic load envelopes with safety factor ≥ 1.5 under worst-case combined loading.

O2. Quantify system-level failure probability as a function of joint efficiency (η_j), segment count (N), and inspection cadence through Monte Carlo simulation—establishing minimum η_j thresholds for 99.9% ten-year system survival.

O3. Validate the modular stress profile against the Edwards & Westling [8] NIAC monolithic baseline.

O4. Develop a lifecycle cost model comparing modular vs. monolithic architectures, quantifying the economic crossover.

O5. Perform material sensitivity analysis across $\sigma_u = 30$–70 GPa, demonstrating robustness to near-term CNT performance.

O6. Map TRL for each subsystem, identifying binding constraints for deployment timeline.

O7. Evaluate repair infrastructure trade space—depot count, placement strategy, repair speed—and identify the binding constraint on MTTR reduction.

1.3 Contributions

2 Literature Review

2.1 Tether Architecture

Early feasibility studies assumed a monolithic, exponentially tapered ribbon [8]. Luo et al. [9] demonstrated a 56% peak-stress reduction using 5–6 optimized segments with connecting platforms, and subsequently analyzed segmented stability [10]. Popescu and Sun [15] introduced a bio-inspired bundle architecture with active self-repair. The IAA Study Group 3.10 recommended a one-meter-wide woven ribbon with in-plane redundancy [18]. An alternative "Spaceline" anchored to the Moon

ID	Contribution
C1	A coupled system-level feasibility assessment of a modular CNT tether
C2	Variable-length mass-equalized segmentation methodology (dual design envelopes)
C3	Monte Carlo reliability surface $P_{\text{sys}}(N, \eta_j, \text{cadence}, p_{\text{det}})$—first published quantification
C4	Minimum viable CNT strength for modular architecture under both tapering philosophies
C5	Lifecycle economic comparison with phased-construction revenue advantage
C6	2D Coriolis-coupled dynamic model with multi-climber resonance separation rule
C7	Repair infrastructure trade-space analysis: inspection cadence as binding MTTR constraint

Table 1: Summary of contributions.

sidesteps the taper problem and is feasible with existing materials [14], though it cannot provide direct surface-to-orbit access.

Gap: No study couples joint reliability with segment geometry, repair logistics, and lifecycle economics at system level.

2.2 CNT Mechanics

Carbon nanotubes remain the reference material: laboratory coupon strengths of 80 GPa [2] and continuous kilometer-scale fibers at 4.1 $\text{N} \cdot \text{tex}^{-1}$ (≈5.3 GPa at $\rho = 1300$ kg/m^3; Niu et al. 12) bracket the assumed $\sigma_u = 50$ GPa baseline. This baseline is aspirational—it has not been demonstrated at ribbon scale—and represents a 10× gap from the best demonstrated continuous-fiber performance. The trajectory of CNT fiber development [11] and defect-tolerant strength scaling analysis [5] suggest that narrowing this gap is plausible but will require fundamental advances in spinning technology and defect control.

2.3 Tether Dynamics

Pendulum periods, climber harmonics, and active damping are well-modeled for continuous tethers [7, 13]. Joint compliance effects on segmented tether stability have not been quantified.

2.4 Deployment and Economics

Edwards' GEO-down bootstrap is established. Lifecycle NPV integrating modular repair costs vs. monolithic replacement has not been computed. Construction phasing—where modular architecture generates revenue incrementally—has not been quantified.

3 Theoretical Foundations

3.1 Taper Profile Derivation

A co-rotating tether element at radial distance r from Earth's center experiences net outward acceleration:

$$a_{\text{net}}(r) = \omega^2 r - \frac{GM}{r^2}, \tag{1}$$

where $\omega = 7.292e-5$ rad/s is Earth's angular velocity and $GM = 3.986e14$ m^3/s^2. At GEO ($r_{\text{GEO}} \approx 42164$ km), $a_{\text{net}} = 0$ by definition.

Source	σ_{design} [GPa]	ρ [kg/m^3]	T_r (pub.)	T_r (calc.)	Stress basis
Edwards & Westling [8]	100	1,300	~1.9	1.88	σ_u, no SF
Pugno [16]	100	1,300	1.9	1.88	σ_u (defect-free)
Popescu & Sun [15]	65	1,325	2.6	2.68	$\sigma_u/2$ (theoretical)
This paper (optimistic)	50	1,300	—	3.52	σ_u, no SF
Aravind [1]	50	1,500	4.28	4.27	$\sigma_u/2$, higher ρ
Pearson [13]	46.5	2,200	~10	9.88	σ_u (graphite)
This paper (conservative)	25	1,300	—	12.40	σ_u/SF
Popescu & Sun [15]	21.5	1,631	36.9	39.4	$\sigma_u/2$ (STR)

Table 2: Reconciliation of published taper ratios. Every value is recoverable from Eq. (3) within 7%.

For a uniform-stress taper, the cross-sectional area satisfies:

$$\frac{\mathrm{d}(\ln A)}{\mathrm{d}r} = -\frac{\rho}{\sigma_{\text{design}}}\, a_{\text{net}}(r). \tag{2}$$

Below GEO, $a_{\text{net}} < 0$ so A increases toward GEO. Above GEO, $a_{\text{net}} > 0$ so A decreases. The maximum area occurs at GEO: $A_{\text{max}} = A_{\text{base}} \times T_r$, where T_r is the taper ratio.

3.2 The Taper Ratio Discrepancy—A First-Class Finding

Published taper ratios span $T_r \approx 1.6$ to $T_r \approx 37$, yet all are derivable from the same integral. The taper ratio $T_r = A(\text{GEO})/A(\text{base})$ is:

$$\ln(T_r) = \frac{\rho}{\sigma_{\text{design}}} \times I_{\text{geo}}, \tag{3}$$

where $I_{\text{geo}} = \int_R^{r_{\text{GEO}}} |a_{\text{net}}(r)|\,\mathrm{d}r \approx 4.84e7$ m^2/s^2 is a fixed geometric constant. The taper ratio depends on exactly two material inputs—density ρ and design stress σ_{design}—plus the often-unstated question of what σ_{design} includes.

Three hidden assumptions determine every published T_r:

1. **Stress basis:** Does σ_{design} equal σ_u (ultimate), σ_u/SF (with safety factor), or $\sigma_u \times \eta_j \times \chi_{\text{rad}}$/SF (full knockdown)?
2. **Material properties:** What (ρ, σ_u) pair is assumed?
3. **Base area sizing:** Is A_{base} for a thin seed ribbon (bootstrap thickening) or a full-payload-capacity ribbon?

Our dual design envelopes are:

Philosophy	σ_{design}	T_r (σ_u=50 GPa)	Interpretation
Optimistic	σ_u	3.52	Shape at full strength; SF for local checks
Conservative	$\sigma_u/\text{SF} = \sigma_{\text{allow}}$	12.40	SF applied to taper geometry
Full knockdown	$\sigma_u \chi_{\text{rad}} \eta_j/\text{SF}$	22.61	All design factors in taper

Table 3: Dual design envelopes.

The optimistic approach yields $N \approx 83$ segments at $m^\star = 18$ t/segment with A_{base} sized for full payload capacity. The conservative approach yields $N \approx 505$ segments and $M_{\text{total}} \approx 9128$ t.

3.3 Specific Strength and Feasibility Boundary

The minimum viable CNT strength is defined as the σ_u at which the architecture "closes"—all segments fit within the 30 t launch cap.

σ_u [GPa]	T_r (opt.)	N (opt.)	M_{total} (opt.)	Feasible
30	8.15	290	5235 t	Yes
35	6.04	191	3443 t	Yes
40	4.82	137	2472 t	Yes
50	3.52	83	1497 t	Yes
60	2.86	58	1043 t	Yes
70	2.46	44	785 t	Yes

Table 4: Material sensitivity under optimistic tapering.

3.4 Variable-Length Mass-Equalized Segmentation

Segment length inversely follows the local ribbon area so that mass per segment is nearly constant:

$$L_j = \frac{m^\star}{\rho\, A_{\text{base}}\, \tilde{A}(y_{j,\text{mid}})}, \tag{4}$$

where $m^\star = 18000$ kg is the target per-segment mass (60% of the 30 t launch allowable).

3.5 Shear-Lap Joint Sizing

A sleeve joint of length ℓ and width b fails in interfacial shear when:

$$\ell_{\min} = \frac{T_j}{2b\,\tau_s}, \tag{5}$$

where $\tau_s = 0.42\,\sigma_{\text{allow}}$. With $T_{\max}$ at GEO and $b = 1.20$ m, this yields $\ell_{\min} = 2.8$ m. The baseline sleeve is sized at $\ell = 3.0$ m, $t_s = 4$ mm, mass = 35 kg.

4 Modular System Architecture

4.1 Baseline Configuration

4.2 Joint Candidates

4.3 Hazard Rate Model

Joint failures are modeled by an Arrhenius law:

$$\lambda_j(T) = \lambda_{0,\text{pre}} \exp\left(-\frac{Q}{k_B T}\right) \left(\frac{0.97}{\eta_j}\right)^4, \tag{6}$$

where $Q = 1.1$ eV is an assumed activation energy for diffusion-controlled void growth [6], and the stress-hazard coupling exponent 4 is adopted from the inverse power law for fatigue life [3].

Parameter	Value	Source
L_{total}	100000 km	GEO + counterweight
ρ	1300 kg/m^3	CNT yarn bulk density
σ_u (baseline)	50 GPa	2030 target; aspirational (§2.2)
SF	2.0	ISO 14692
η_j (baseline)	0.95	Assumed; full range $\eta_j \in [0.70, 0.97]$ explored
$m^\star$	18000 kg	Target per-segment mass
m_{launch}	30000 kg	Starship-class GEO injection
m_{climber}	20000 kg	Construction/cargo climber
v_{climber}	150 m/s	Transit velocity
m_{cw}	600000 kg	Edwards & Westling [8]

Table 5: Baseline design parameters.

Joint Type	η_j	Mass [kg]	Install [h]	10-yr P_{fail}
Bolted collet	0.88	95	2.1	6.6×10^{-3}
Nanobond sleeve (baseline)	0.95	35	1.4	4.6×10^{-3}
Sleeve + BNNT overplate	0.94	38	1.8	4.7×10^{-3}

Table 6: Joint candidates. Properties are notional design targets; no experimental full-scale CNT sleeve coupler data exists (§8).

Volume scaling. Full-scale sleeves (3.0 m × 1.2 m × 4 mm) are 6,000× the volume of test coupons. Weibull weakest-link scaling with modulus $m = 6$ gives:

$$\lambda_{\text{full}} = \lambda_{\text{coupon}} \times \left(\frac{V_{\text{sleeve}}}{V_{\text{coupon}}}\right)^{1/m} = 1.2 \times 10^{-8} \times 6000^{1/6} = 5.12e-8 \text{ h}^{-1}. \tag{7}$$

We sweep $m \in \{2.7, 4, 6, 8, 10\}$ to bound uncertainty, noting that Weibull scaling for time-dependent hazard rates is an engineering approximation [4, 17].

4.4 Thermal Profile

Zone	Altitude	T [K]	Dominant Environment
1	0–200 km	250	Atmospheric drag, eclipse cycling
2	200 km–35786 km	280	LEO–GEO corridor
3	> 35786 km	300	Cislunar, sustained solar exposure

Table 7: Three-zone thermal profile.

5 Simulation Methodology

5.1 Static Taper Validation

Numerical integration of Eq. (2) at 1 km resolution (100001 points). Stress uniformity $\sigma_{\text{std}}/\sigma_{\text{mean}} \approx 10^{-17}$ (machine precision) confirms correct implementation. Sensitivity sweep across $\sigma_u \in \{30, 35, 40, 50, 60, 70\}$ GPa at both design envelopes.

5.2 Monte Carlo Joint Reliability

5.2.1 Exponential Baseline

Parameter space: 2,268 combinations—$N \in \{12, 15, 18, 21, 24, 50, 100, 200, 500\}$, $\eta_j \in \{0.70$–$0.97\}$, inspection cadence $\in \{1, 2, 5, 10\}$ passages, $p_{\text{det}} \in \{0.50$–$0.995\}$. Per trajectory: (1) draw time-to-failure for each joint from $\text{Exp}(\lambda_j(T_{\text{local}}))$; (2) step through inspection epochs; (3) cascade check—redistribute load from failed joints to neighbors; two adjacent unrepaired failures trigger immediate system loss; (4) detection roll (p_{det}); (5) detected joints are repaired. 10^5 trajectories per combination (2.27×10^8 total).

5.2.2 Weibull Age-Dependent Extension

The failure model is extended to the two-parameter Weibull distribution:

$$h(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1}, \tag{8}$$

where $\beta > 1$ gives increasing hazard (wear-out). Expanded to 12,600 combinations adding $\beta \in \{1.0, 1.3, 1.5, 2.0, 2.5\}$. Each joint maintains cumulative hazard state, and stress redistribution triggers remaining-life recalculation. Validation: $\beta = 1.0$ recovers the exponential model at identical seeds.

5.2.3 Hazard Rate Sensitivity

One-at-a-time perturbation ($\pm 20\%$, $\pm 40\%$) of seven parameters: Q, Weibull modulus m, zone temperatures (T_1, T_2, T_3), coupon rate λ_{coupon}, and volume ratio V_{ratio}.

5.3 Dynamic Modal Analysis

5.3.1 1D Baseline

For a uniform-stress fixed-free string under gravity-gradient tension:

$$c = \sqrt{\sigma_{\text{design}}/\rho}, \quad T_{1,\text{pendulum}} = 4L/c. \tag{9}$$

At $\sigma_{\text{allow}} = 25$ GPa: $c \approx 4385$ m/s, $T_1 \approx 25.3$ h. A 500-node lumped-mass-spring chain confirms joint compliance ($\eta_j = 0.95$ vs. 1.0) shifts the elastic fundamental frequency by 2.3%.

5.3.2 2D Coriolis-Coupled Finite Element Model

A 2D model in the rotating frame. Each of 500 nodes carries two DOFs: longitudinal u (radial) and transverse v (in-plane), giving a 998-DOF system.

Element formulation: Elastic (longitudinal EA/L, reduced by η_j at joints), tension (transverse $T(r)/L$), and gravity-gradient ($\omega^2 + 2GM/r^3$) stiffness contributions. Coriolis coupling via skew-symmetric gyroscopic matrix $\mathbf{G}$.

Damping: Rayleigh damping calibrated to $\zeta = 0.01$ at the first two modes, consistent with the loss tangent ~0.045 for dry-spun CNT fibers [19].

Time integration: Newmark-β ($\beta = 0.25$, $\gamma = 0.5$) with $\Delta t = 500$ s. Non-uniform mesh with Gaussian refinement near GEO and 600000 kg counterweight at the free tip.

Validation: Six checks pass: $\mathbf{K}$ positive definite, 1D longitudinal recovery (2.4% error), Coriolis matrix skew-symmetry ($\|\mathbf{G}+\mathbf{G}^\top\|/\|\mathbf{G}\| = 0$), energy conservation ($< 0.1\%$ drift over 100 h), mesh convergence ($< 0.1\%$ between 250 and 500 elements), and negligible joint compliance effect on transverse modes.

5.3.3 Single-Climber and Multi-Climber Simulations

A 20 t climber at 150 m/s (transit 185 h) applies gravity-gradient load (longitudinal, up to 196 kN) and Coriolis force (transverse, ∼438 N). Multi-climber resonance sweep: departure separations 10–50 h, 4 climbers, measuring peak transverse displacement and tension perturbation.

5.4 Lifecycle Cost Model

Modular:

$$\mathrm{NPV} = \sum_{t=1}^{30} \frac{R(t)\, f_{\mathrm{op}}(t) - C_{\mathrm{build}}(t) - C_{\mathrm{ops}}(t) - C_{\mathrm{repair}}(t)}{(1+r)^t}, \tag{10}$$

where $f_{\mathrm{op}}(t) = \max(0, \min(1, (n_{\mathrm{deployed}}/N - 0.6)/0.4))$—revenue ramps from 0 at 60% completion to full at 100%, assuming a GEO-outward/inward deployment strategy [8].

Monolithic: Zero revenue until 100% complete; simplified annual P_{fail}.

Sweep: launch cost $\{500, 1000, 1500, 2000\}$ \$/kg × discount rate $\{5, 7, 10\}\%$ × payload revenue $\{200, 300, 500\}$ \$/kg.

5.5 Repair Infrastructure Trade Space

Analytical MTTR model:

$$\mathrm{MTTR} = t_{\mathrm{wait}} + t_{\mathrm{travel}} + t_{\mathrm{repair}}, \tag{11}$$

with $t_{\mathrm{wait}} \sim \mathrm{Unif}[0, \mathrm{cadence} \times L_{\mathrm{total}}/v_{\mathrm{climber}}]$. Three depot placement strategies evaluated (uniform, joint-density-weighted, hybrid) across $n_{\mathrm{depots}} \in \{0, 1, \ldots, 50\}$ and $v_{\mathrm{repair}} \in \{150, 300, 600\}$ m/s.

6 Results

6.1 Taper Profile Validation

$T_r = 12.40$ at $\sigma_{\mathrm{allow}} = 25$ GPa, $T_r = 3.52$ at $\sigma_u = 50$ GPa. Stress uniformity: $\sigma_{\mathrm{std}}/\sigma_{\mathrm{mean}} = 5.39 \times 10^{-17}$ (machine precision).

6.2 Dual Design Envelope

Under **optimistic tapering** (σ_u, no SF on shape): $T_r = 3.52$, $N \approx 83$ segments, $M_{\mathrm{total}} \approx 1497$ t. Under **conservative tapering** (σ_{allow}): $T_r = 12.40$, $N \approx 505$ segments, $M_{\mathrm{total}} \approx 9128$ t—a 6× increase in segments and mass.

Edwards & Westling's lower segment count ($N \approx 18$) reflects their seed-ribbon A_{base}. Our $A_{\mathrm{base}} = m_{\mathrm{climber}} \times g/\sigma_{\mathrm{design}}$ gives a wider ribbon supporting a 20 t climber from day one.

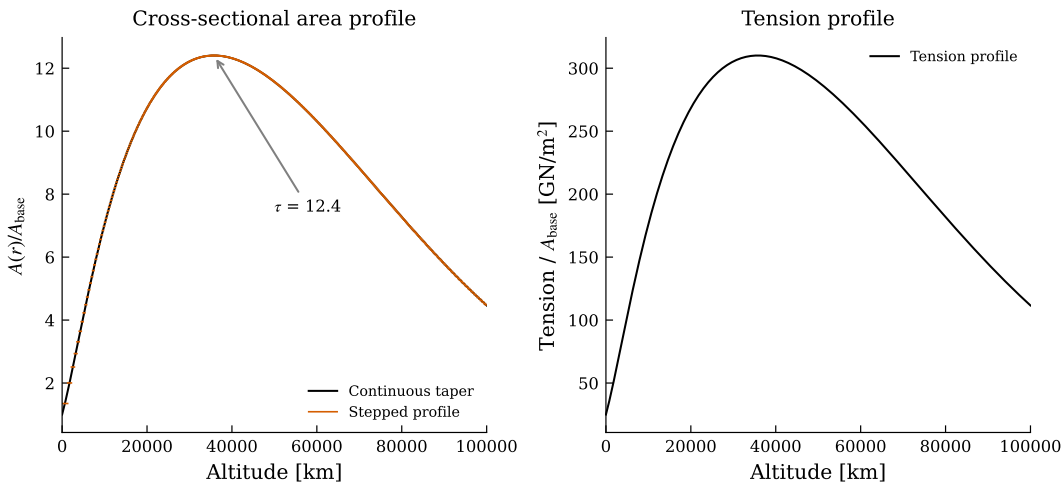

Figure 1: Continuous vs. stepped taper profile $A(r)$ and $T(r)$.

6.3 Monte Carlo Reliability Surface

6.3.1 Exponential Baseline

The 4D sweep (2,268 combinations $\times$ 10^5 trajectories) reveals:

Well-designed regime ($\eta_j \geq 0.88$, $p_{\text{det}} \geq 0.90$, $N \leq 24$): $P_{\text{sys}} > 99.5\%$ (conditional on $Q \approx 1.1$ eV).

Degraded regime ($\eta_j < 0.80$, $p_{\text{det}} < 0.70$, or $N > 100$): P_{sys} drops to 72.7% at worst case. The 99.9% contour requires $\eta_j \geq 0.88$ with $p_{\text{det}} \geq 0.90$; the 99.0% contour requires $\eta_j \geq 0.80$ with $p_{\text{det}} \geq 0.70$.

6.3.2 Weibull Extension

At the baseline configuration (N=83, η_j=0.95, cadence=1, p_{det}=0.995):

β	P_{sys}	Mean repairs	Median MTTR [h]
1.0 (exponential)	0.99995	0.41	218
1.5	0.99999	0.034	219
2.0 (wear-out)	1.00000	0.003	212
2.5 (strong wear-out)	1.00000	0.0003	207

Table 8: Weibull extension results at baseline design point.

6.3.3 Q-Sensitivity

A targeted sweep at $Q = \{0.8, 1.1, 1.4\}$ eV across 270 combinations $\times$ 10^5 trajectories reveals stark results: at $Q = 0.8$ eV, $P_{\text{sys}} = 0$ everywhere; at $Q = 1.4$ eV, $P_{\text{sys}} = 1.000$ everywhere. The entire well-designed regime exists within a narrow activation-energy window.

Hazard rate sensitivity. One-at-a-time perturbation identifies Q as the overwhelmingly dominant parameter ($\sim$7 orders of magnitude per $\pm$40%). Zone 3 temperature ranks second ($\sim$3 orders). Remaining parameters contribute $\leq$0.5 decades.

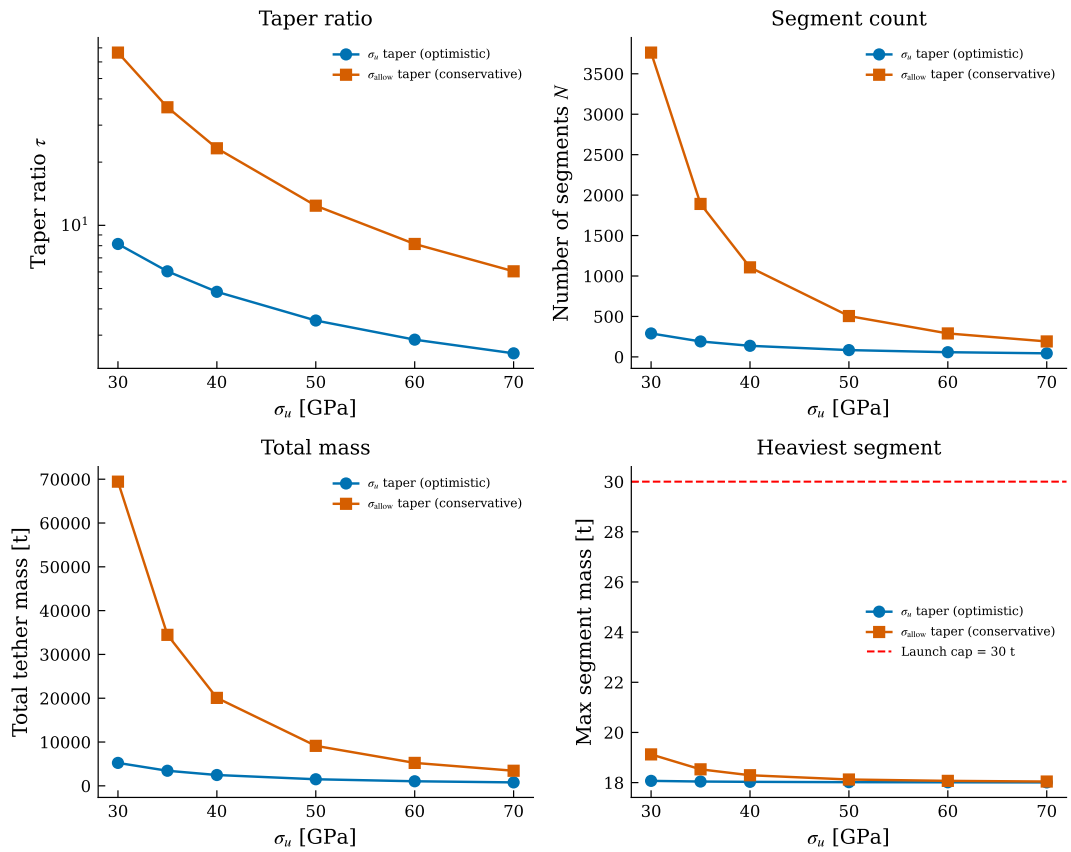

Figure 2: Dual design envelope: T_r, N, M_{total}, $m_{j,\max}$ vs. σ_u for both philosophies.

6.4 MTTR Distribution and Depot Infrastructure

From ∼202.7 million repair events: median MTTR ∼227 h (cadence=1), ∼1275 h (cadence=10). MTTR is dominated by travel time (150 m/s over up to 100000 km).

Depots	v_{repair} [m/s]	E[MTTR] [h]	$P(\text{MTTR} \leq 72\text{ h})$
0	150	183	5.8%
5	150	102	33.6%
10	150	98	35.8%
50	150	95	37.6%
5	600	96	37.0%
50	600	94	38.0%

Table 9: Depot infrastructure analysis (N=83, optimistic taper). $P(\text{MTTR} \leq 72\text{ h})$ saturates near 38%.

Key finding: $P(\text{MTTR} \leq 72\text{ h})$ saturates near 38% regardless of depot count or repair speed. The binding constraint is inspection cadence: a single traversal at 150 m/s takes 185 h, so the expected wait time of 93 h already exceeds 72 h.

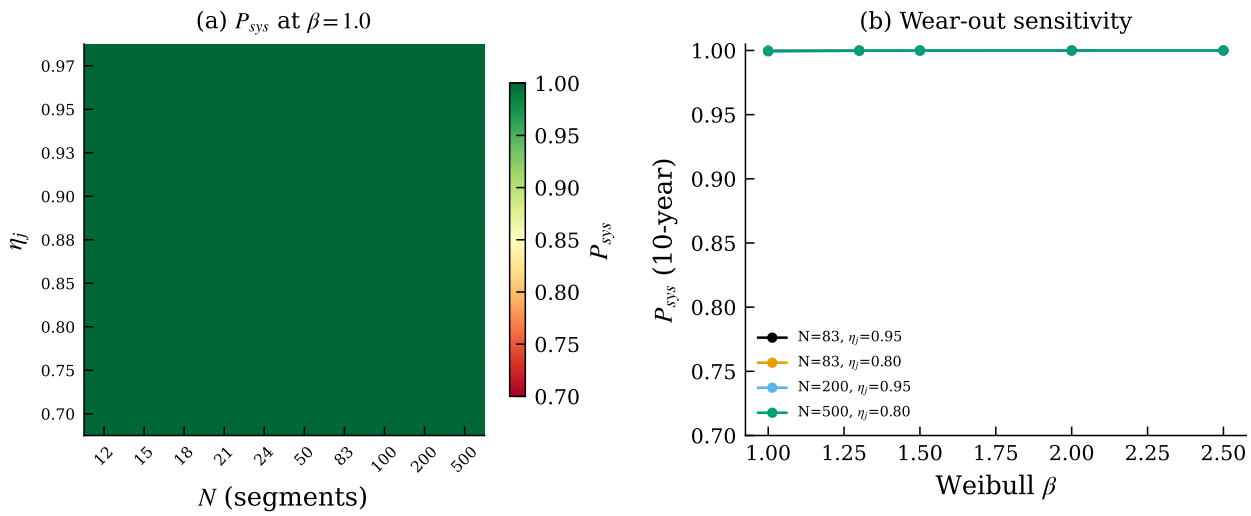

Figure 3: (a) $P_{\text{sys}}(N, \eta_j)$ heatmap at $\beta = 1.0$; (b) P_{sys} vs. Weibull β at four design points.

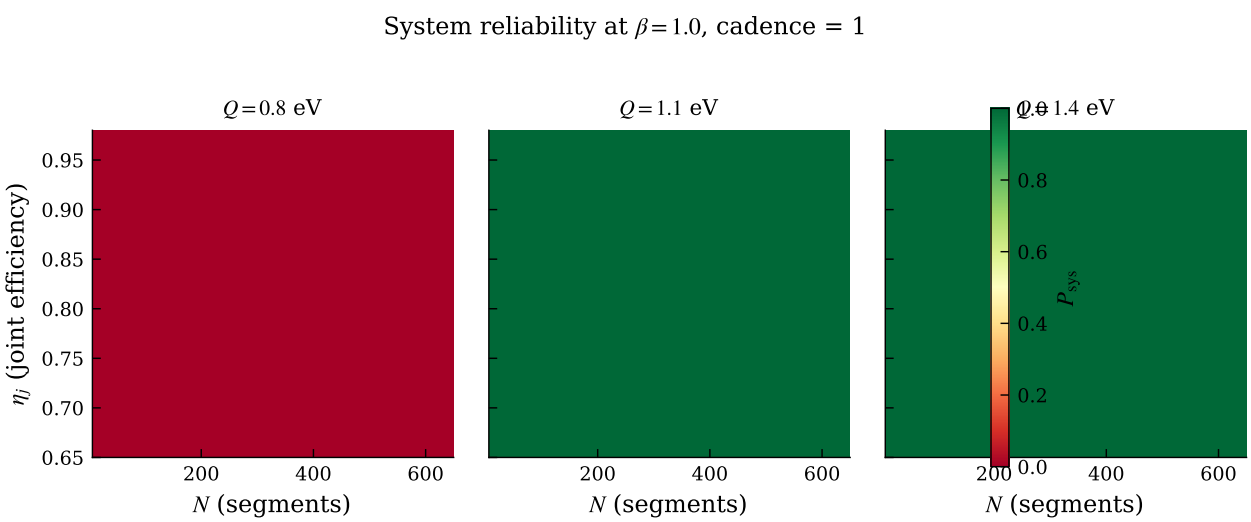

Figure 4: $P_{\text{sys}}(N, \eta_j)$ heatmaps at $Q = \{0.8, 1.1, 1.4\}$ eV (3-panel).

6.5 Dynamic Modal Analysis

6.5.1 2D Modal Results

6.5.2 Multi-Climber Resonance

A clear resonance peak at 35 h separation (near $T_{1,\text{trans}} = 34.8$ h): peak transverse displacement reaches 489 km (3.6× single-climber). Off-resonant separations (e.g., 18 h) produce only 129 km.

Design rule: Climber departures within ±5 h of ∼35 h should be avoided.

6.6 Lifecycle Cost Analysis

Modular advantage: Revenue begins at ∼60% tether completion (year ∼4), ∼2.8 years of partial revenue before monolithic achieves 100%. Modular NPV exceeds monolithic at year 5 across all tested scenarios. The advantage is robust to launch cost ($500–2000/kg), insensitive to revenue

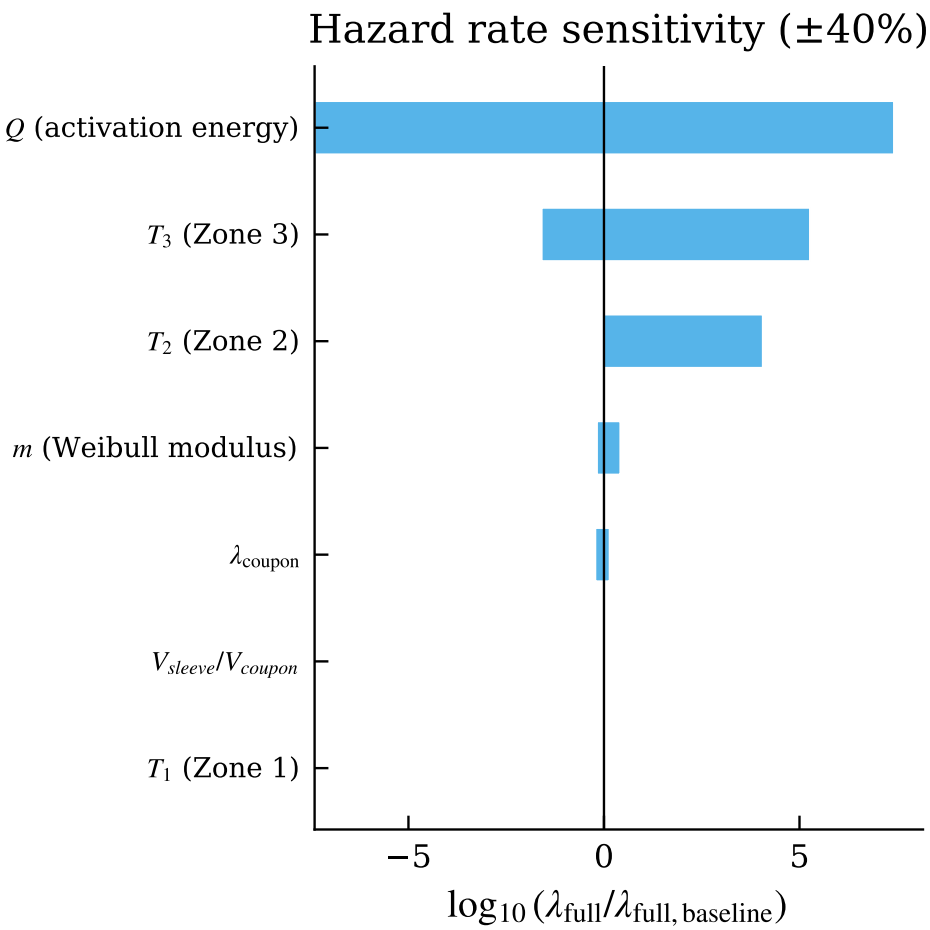

Figure 5: Hazard rate sensitivity tornado diagram—Q dominates.

pricing, and dominated by phased construction rather than repair savings.

Sensitivity: Sweeping $P_{\text{fail,mono}} \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$/yr, the modular NPV advantage persists across all values including the most optimistic monolithic reliability (10^{-4}/yr), because the phased-construction revenue advantage dominates. Sweeping $f_{\text{threshold}} \in \{0.5\text{–}0.9\}$: advantage narrows from \$0.41B to \$0.17B but remains positive everywhere.

7 Discussion

7.1 Taper Ratio: Implications for the Field

The reconciliation in §3.2 shows that published taper ratios spanning $T_r = 1.6$ to $T_r = 37$ reduce to a single formula once three hidden assumptions are identified. Claims that "the taper ratio is ∼2" or "∼10" are not contradictory—they reflect different, usually unstated, design philosophies. We recommend that all future taper analyses explicitly state the stress basis, material density, and A_{base} sizing philosophy.

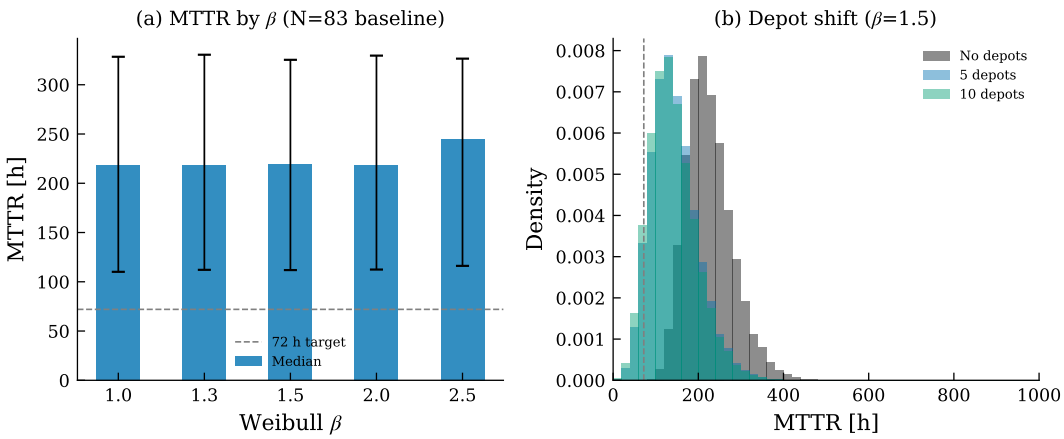

Figure 6: (a) MTTR by Weibull β; (b) depot shift of MTTR distribution.

Property	No Coriolis	With Coriolis	Shift
T_1 transverse [h]	30.0	34.8	+16%
T_2 transverse [h]	10.2	10.5	+2.5%
T_1 longitudinal [h]	10.22	10.46	+2.3%

Table 10: 2D modal analysis results. Coriolis coupling shifts the transverse fundamental by 16%.

7.2 Reliability Under Wear-Out

$P_{\text{sys}} > 99.5\%$ across the well-designed regime persists under Weibull wear-out up to $\beta = 2.5$ (conditional on $Q \approx 1.1$ eV). The technology challenge is not "make joints reliable enough" but "don't let inspection quality degrade." The reliability cliff at $p_{\text{det}} < 0.70$ provides a clear technology target: the UT inspection system must achieve $> 90\%$ single-pass detection probability.

7.3 The Q Problem

The most important finding regarding what remains unknown. The Arrhenius model's exponential dependence on Q/k_BT means that even modest uncertainty (± 0.3 eV) spans the entire range from guaranteed failure to guaranteed success. No amount of engineering optimization can compensate for the wrong Q. This reframes the problem from systems engineering to materials characterization: the first experimental measurement of void-growth activation energy in CNT nano-solder bonds will determine viability.

The 10-orders-of-magnitude swing across $Q \in \{0.8, 1.1, 1.4\}$ eV is not a model limitation but a physical reality of Arrhenius kinetics at 250–300 K. All reliability predictions in this paper are conditional on $Q \in [1.0, 1.2]$ eV.

7.4 The 72 h Target: An Inspection Problem

The Edwards & Westling 72 h repair target is structurally constrained by inspection cadence. With $v_{\text{climber}} = 150$ m/s, a traversal takes 185 h; the expected wait of 93 h already exceeds 72 h. Three paths exist: (a) faster climbers (≥ 300 m/s), (b) continuous SHM, or (c) multiple simultaneous inspection climbers.

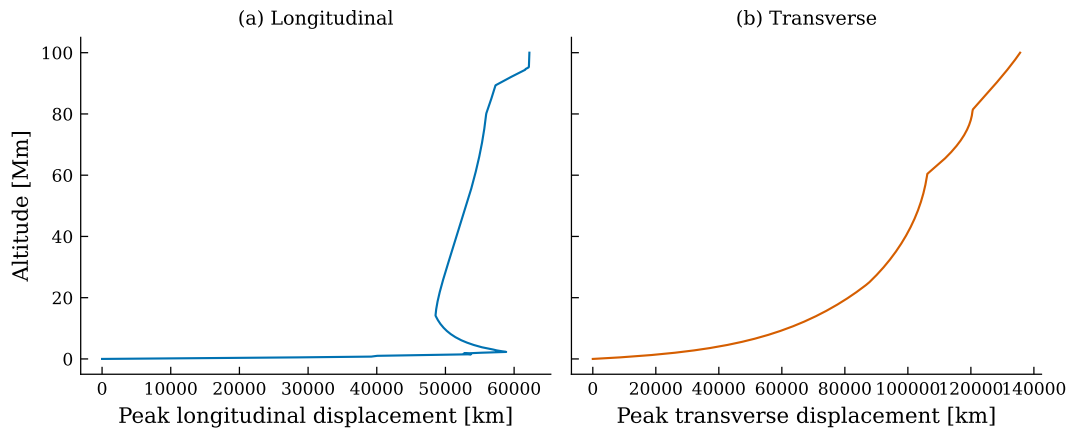

Figure 7: (a) Longitudinal and (b) transverse displacement envelopes vs. altitude for single-climber transit.

Parameter	Value
Tether mass (opt., N=83)	1497 t
Build cost ($1000/kg)	$1.58B
Annual operations	$32M (2% of build)
Max annual revenue	$300M (50 × 20 t × $300/kg)
Repair cost per event	$27.1M
Construction cadence	12 segments/year (∼7 yr build)

Table 11: Lifecycle cost baseline assumptions.

7.5 Comparison to Prior Work

Paper	What They Did	What We Add
Edwards & Westling [8]	Monolithic taper (implicitly at σ_u)	Explicit dual-envelope analysis
Luo et al. [9]	56% stress reduction	Coupled reliability, cascade model, cost
Popescu & Sun [15]	Bio-inspired repair concept	Quantified MTTR, phased economics
Aravind [1]	Canonical taper derivation	System-level integration

Table 12: Comparison to prior work.

7.6 Limitations

1. **2D model:** Omits out-of-plane dynamics, torsion, and ribbon effects. The transverse gravity-gradient body force is omitted; tension-string stiffness exceeds it by >3,000× outside GEO ± 1000 km.

2. **Thermal model:** Three discrete zones rather than continuous thermal profile.

3. **Joint failure:** Actual wear-out exponent for CNT sleeve bonds is unknown. Extreme Q-sensitivity (∼7 orders per ±40%) means substantial uncertainty.

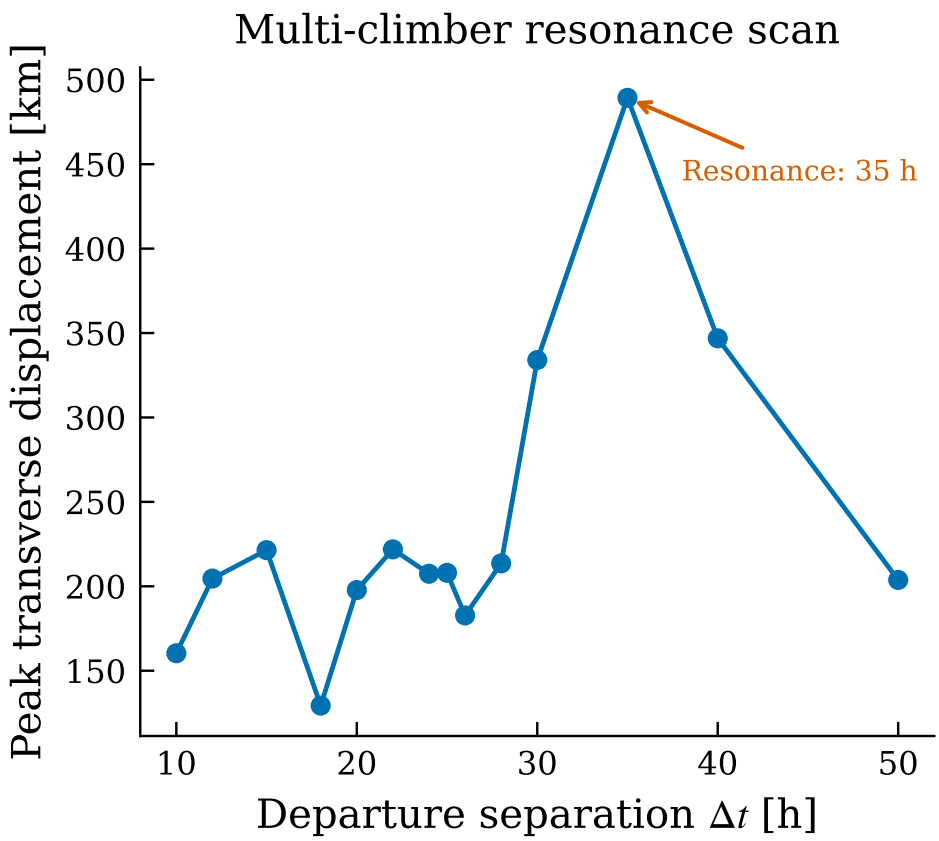

Figure 8: Multi-climber resonance scan: peak transverse displacement vs. departure separation.

4. **No experimental validation:** Every physical parameter is assumed. Experimental next steps in §8.
5. **Monolithic failure model:** Simplified annual probability; asymmetric treatment addressed via sensitivity sweep (§6).
6. **Damping uncertainty:** Baseline $\zeta = 0.01$ is conservative; resonant displacement varies 1.8× across $\zeta = 0.001$–0.05.
7. **Depot cost model:** Parametric (order-of-magnitude) estimates.

8 Technology Gaps and Roadmap

The binding constraint is CNT ribbon production at $\sigma_u \geq 40$ GPa on kilometer-per-day lines. Laboratory coupons reach 80 GPa [2]; continuous fibers achieve ≈5.3 GPa [12]. Defect-tolerant scaling [5] predicts ∼10 GPa macroscopic strength, suggesting a 5× remaining gap.

Continuous SHM is the second critical technology: periodic inspection produces a wait-time floor (93 h) exceeding the 72 h target.

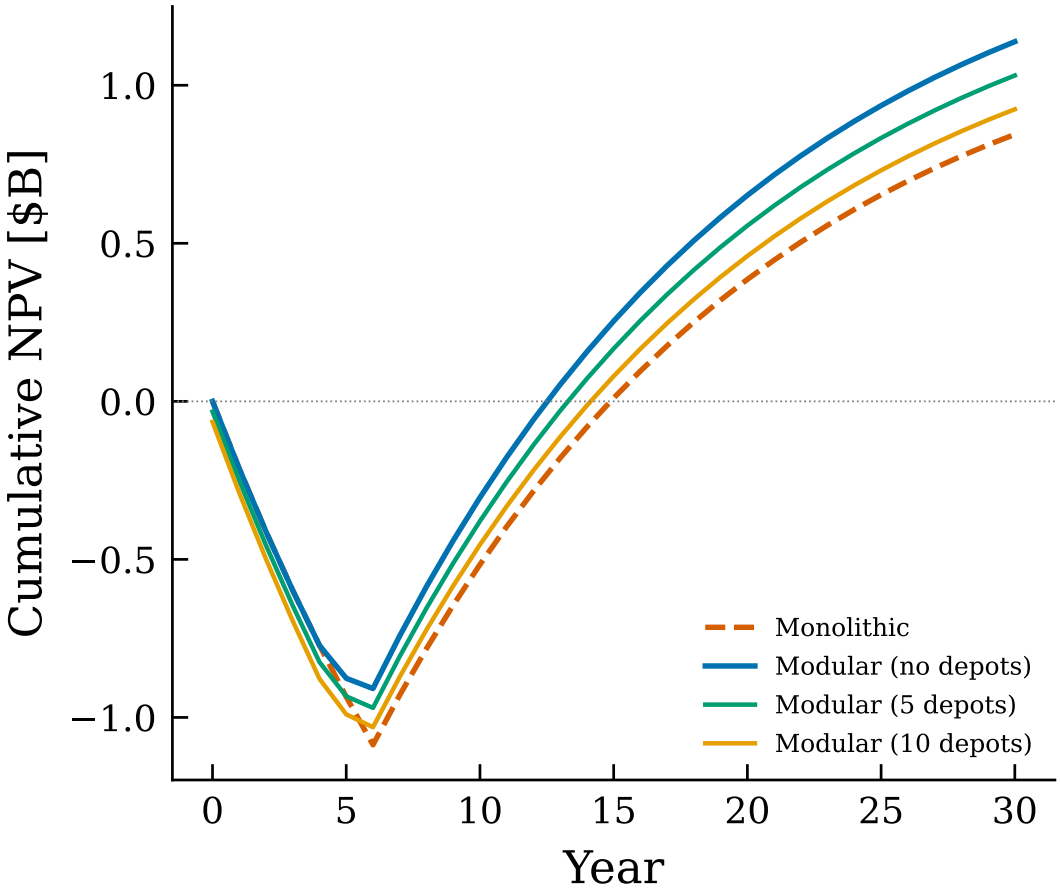

Figure 9: Cumulative NPV: monolithic vs. modular with 0/5/10 depots.

9 Conclusions

This paper presents a coupled system-level analysis of a modular CNT space-elevator tether integrating taper geometry, joint reliability, repair logistics, dynamic stability, and lifecycle economics. Key findings:

C1—Coupled feasibility: The modular architecture is feasible under both design envelopes, with $N \approx 83$ (optimistic) to $N \approx 505$ (conservative) segments at $\sigma_u = 50$ GPa.

C2—Taper ratio resolved: All published values ($T_r \approx 1.6$–37) are recoverable from a single integral once stress basis, density, and base-area sizing are specified.

C3—Reliability surface: $P_{\text{sys}} > 99.5\%$ in the well-designed regime conditional on $Q \approx 1.1$ eV. A Q-sweep reveals 10 orders of magnitude hazard-rate variation across $Q \in \{0.8, 1.1, 1.4\}$ eV. Experimental Q characterization is the binding prerequisite.

C4—Minimum viable strength: Architecture closes at $\sigma_u = 30$ GPa (optimistic). Current continuous fibers achieve ≈5.3 GPa; the 50 GPa baseline remains aspirational.

C5—Economic advantage: Modular outperforms monolithic across all tested cost scenarios, driven by phased-construction revenue (∼60% completion → year 4 of 7). Advantage persists across all $f_{\text{threshold}} \leq 0.9$ and $P_{\text{fail,mono}} \geq 10^{-4}$/yr.

C6—2D dynamics: Coriolis coupling shifts the transverse fundamental by 16% ($T_1 = 34.8$ h). Multi-climber resonance at 35 h separation amplifies transverse displacement 3.6×. Design rule: avoid departures within ±5 h of 35 h.

Subsystem	TRL	Req.	Binding Constraint
CNT ribbon ($\sigma_u \geq$ 50 GPa, km)	3–4	6	Spinning rate × defect density
Nanobond sleeve ($\eta_j \geq$ 0.95)	4	6	Full-scale vacuum bonding
UT inspection ($p_{\text{det}} \geq$ 0.90)	5	7	Climber integration, speed
Robotic sleeve replacement	3	6	Autonomous GEO operations
Climber power beaming	4	6	Atmospheric propagation
Counterweight capture	2–3	5	NEA guidance or mass delivery
Continuous SHM	3	6	100000 km distributed sensing

Table 13: TRL assessment. The binding constraint is CNT ribbon production at $\sigma_u \geq 40$ GPa.

C7—Repair infrastructure: Depots reduce expected MTTR from 183 h to ∼95 h, but $P(\text{MTTR} \leq 72 \text{ h})$ saturates near 38%. Inspection cadence—not depot count—is the binding constraint.

The modular tether reframes the space-elevator challenge from materials perfection to systems engineering and maintainability.

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A Supplementary Figures

Author Contributions

K. Egan: Conceptualization, Methodology, Software, Formal Analysis, Writing—Original Draft, Visualization.

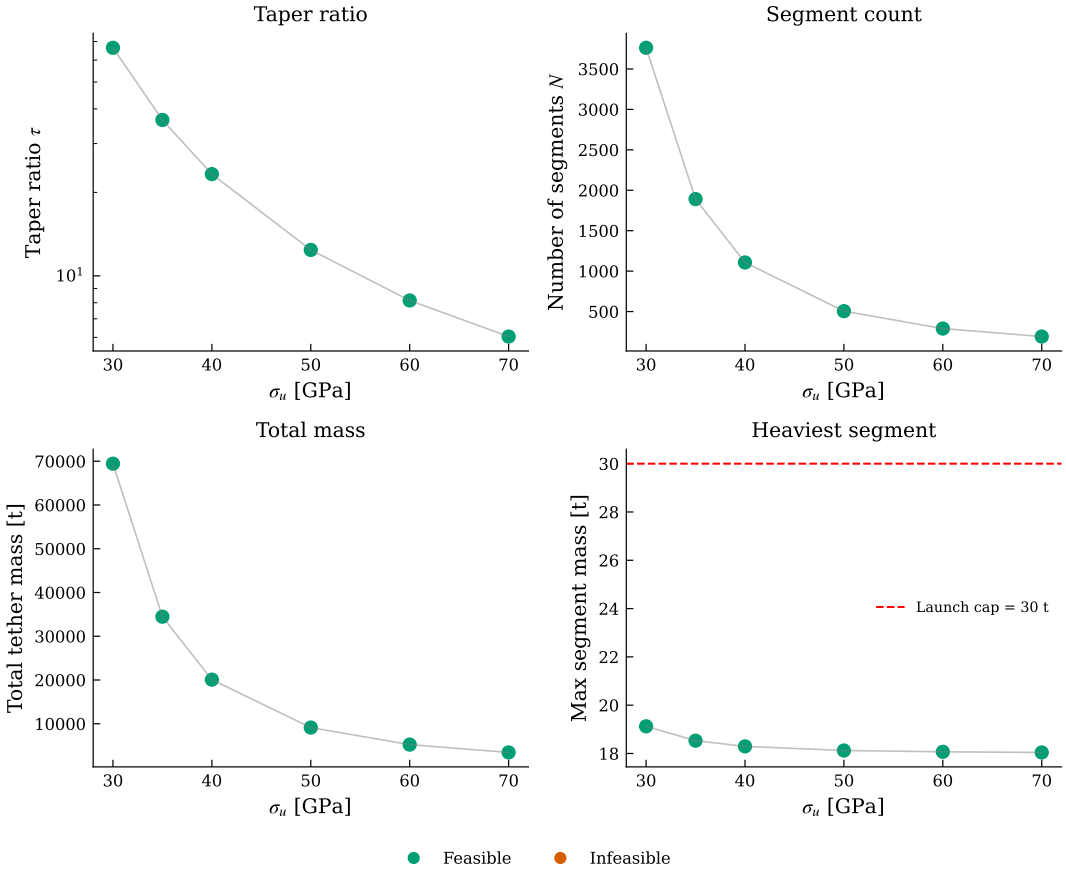

Figure 10: Material sensitivity: taper ratio and mass vs. σ_u.

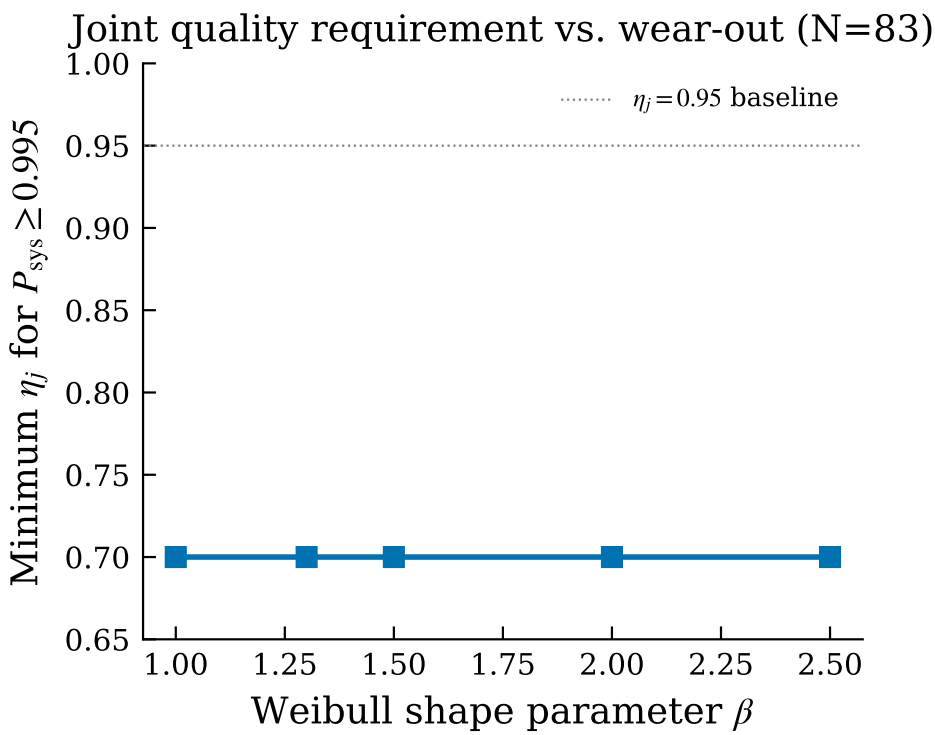

Figure 11: Minimum η_j for $P_{\text{sys}} \geq 0.995$ vs. Weibull β (N=83).

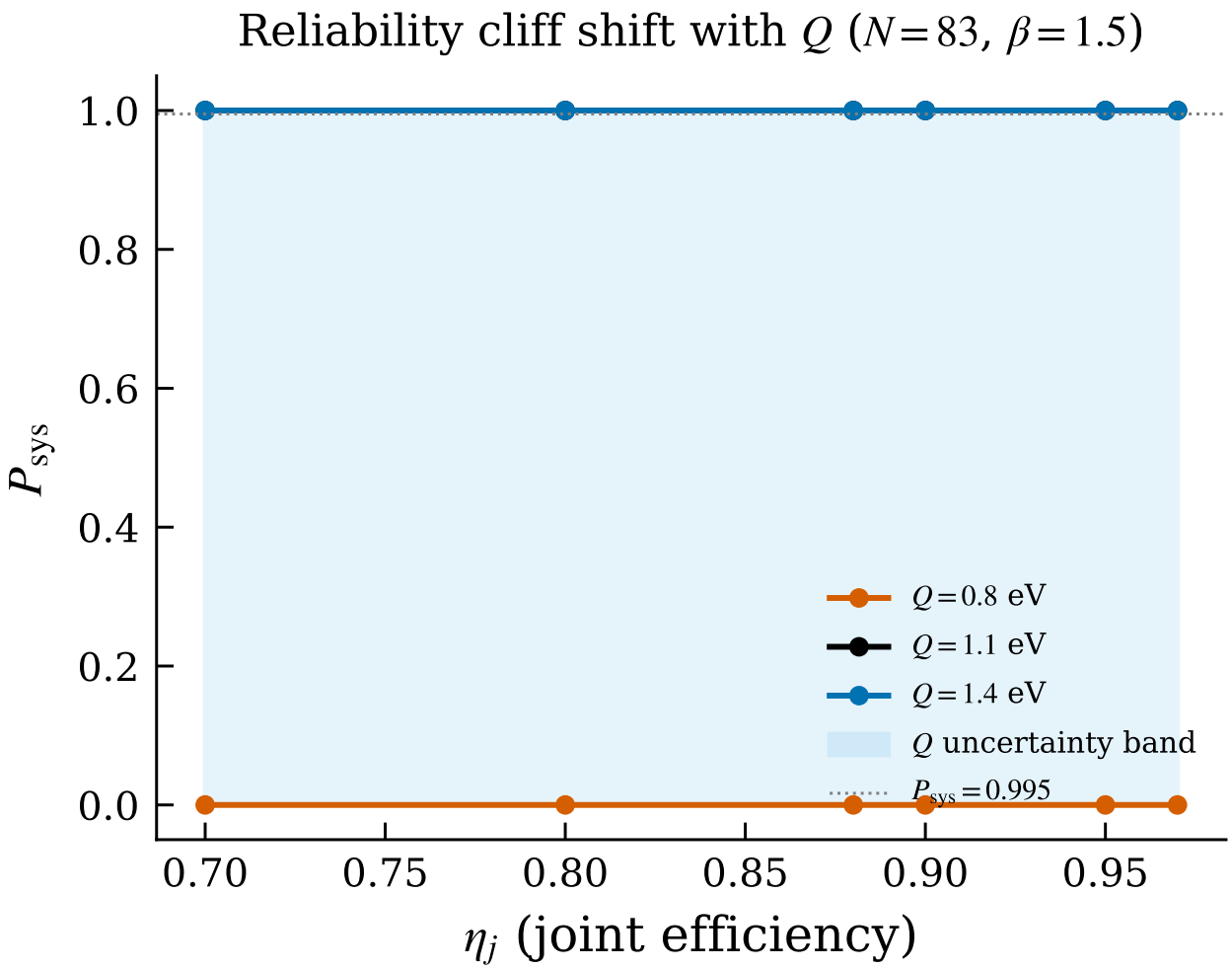

Figure 12: P_{sys} vs. η_j showing Q uncertainty band (N=83, β=1.5).

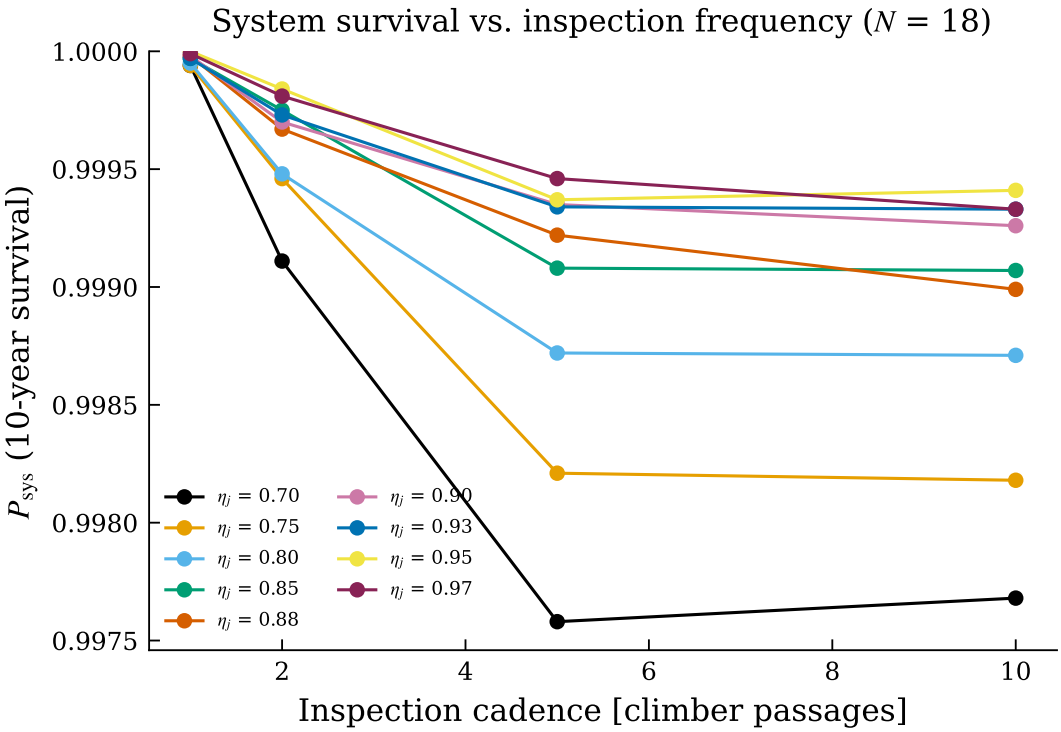

Figure 13: System survival vs. inspection frequency.

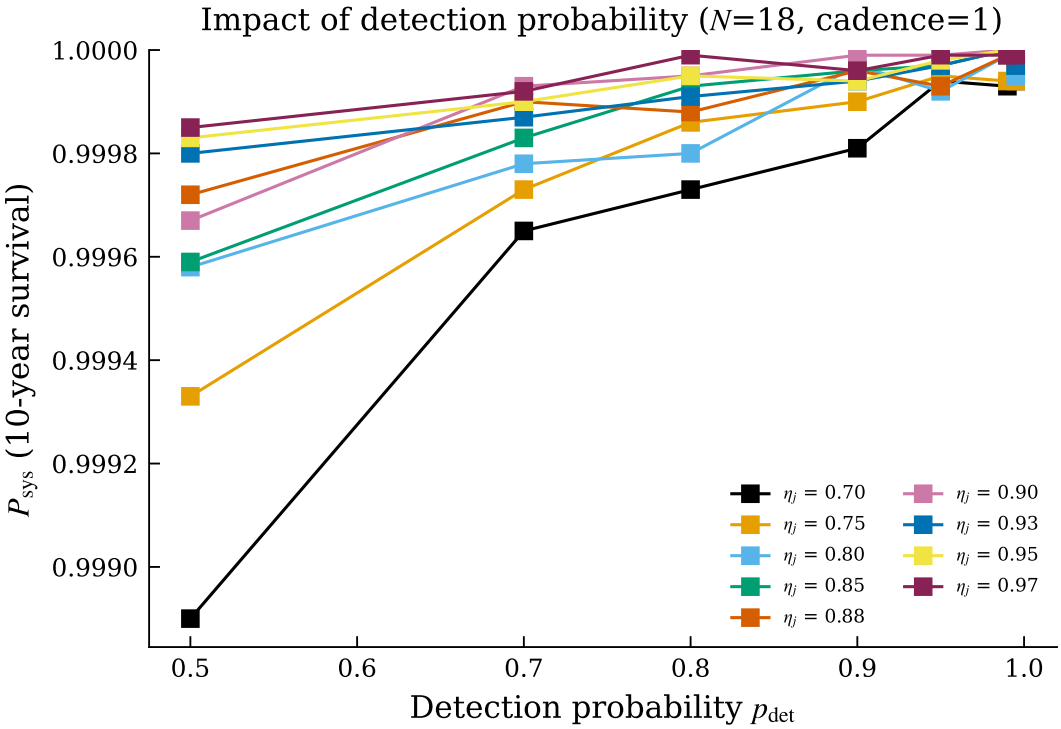

Figure 14: P_{sys} vs. detection probability.

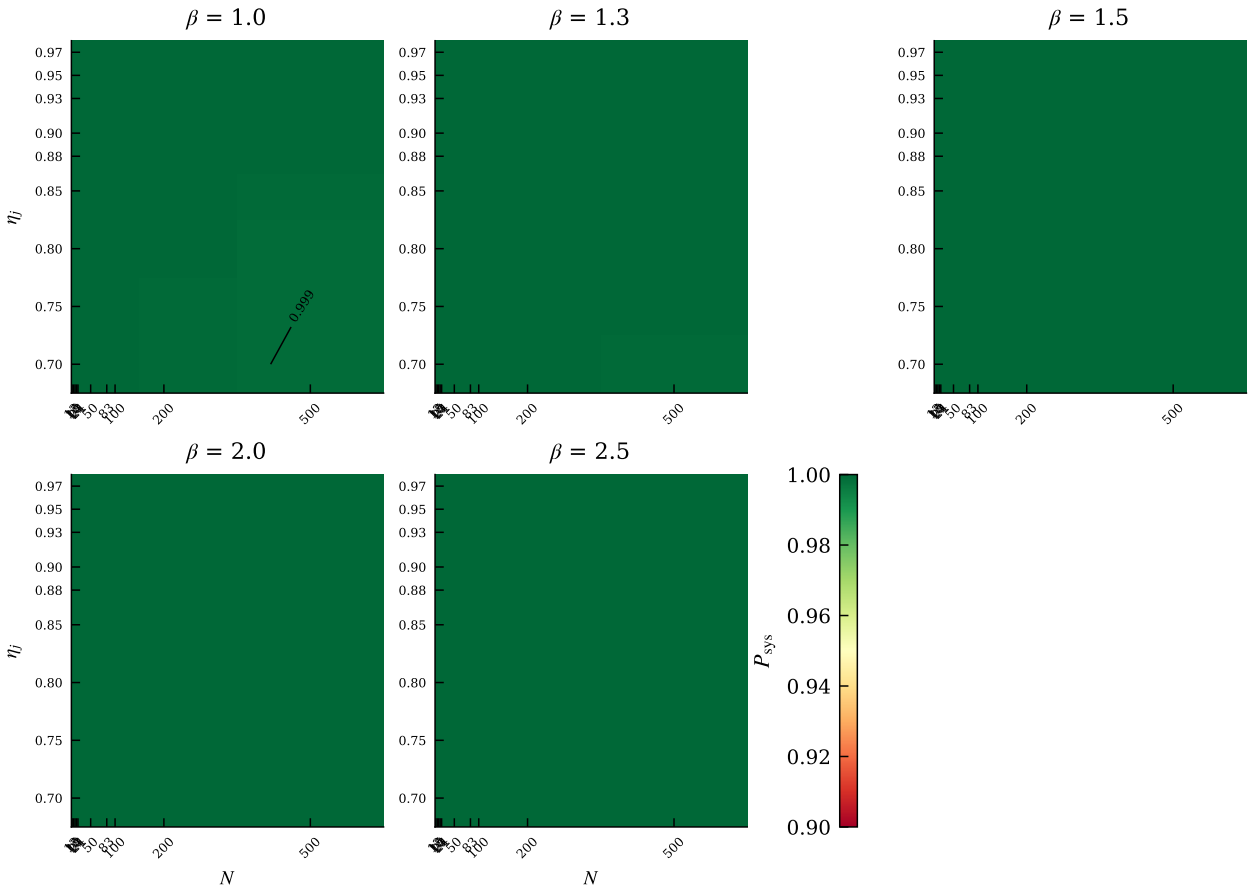

Figure 15: $P_{\text{sys}}(N, \eta_j)$ heatmaps at each β (5 panels).

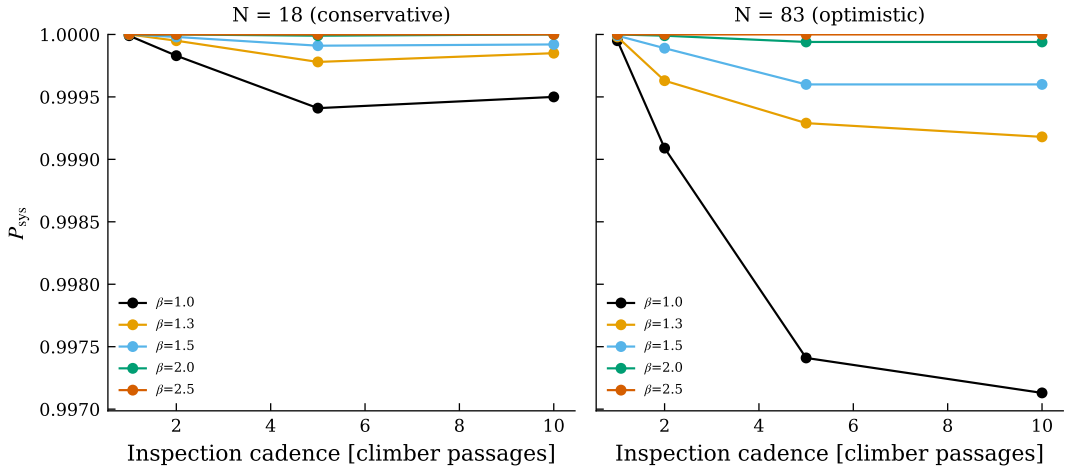

Figure 16: Inspection cadence sensitivity under wear-out.

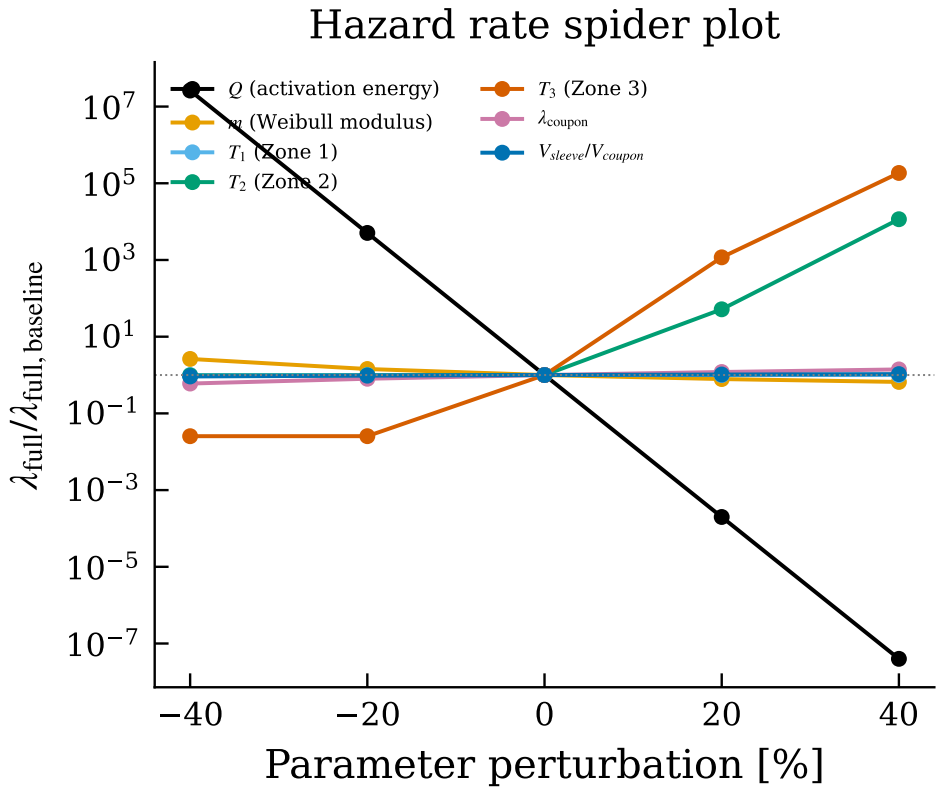

Figure 17: Hazard rate spider plot.

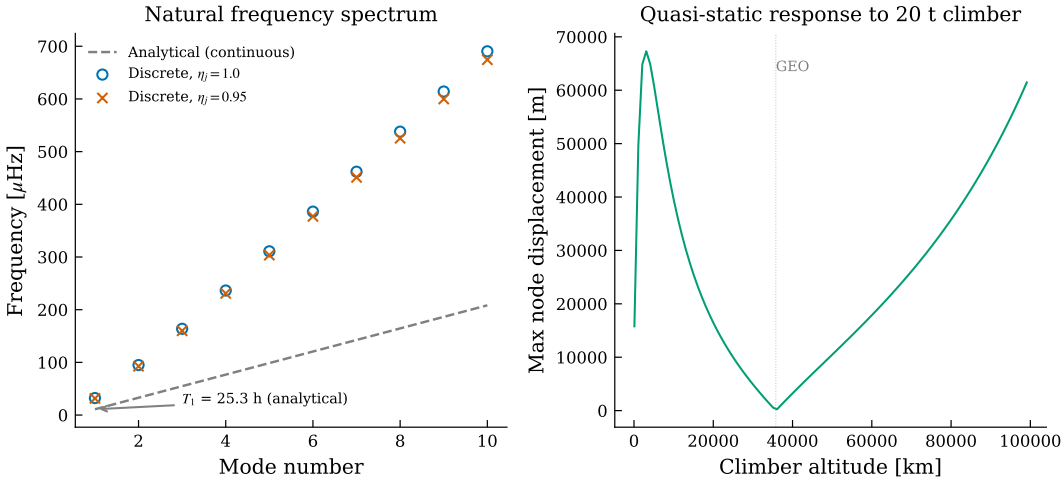

Figure 18: 1D modal frequency spectrum and quasi-static forced response.

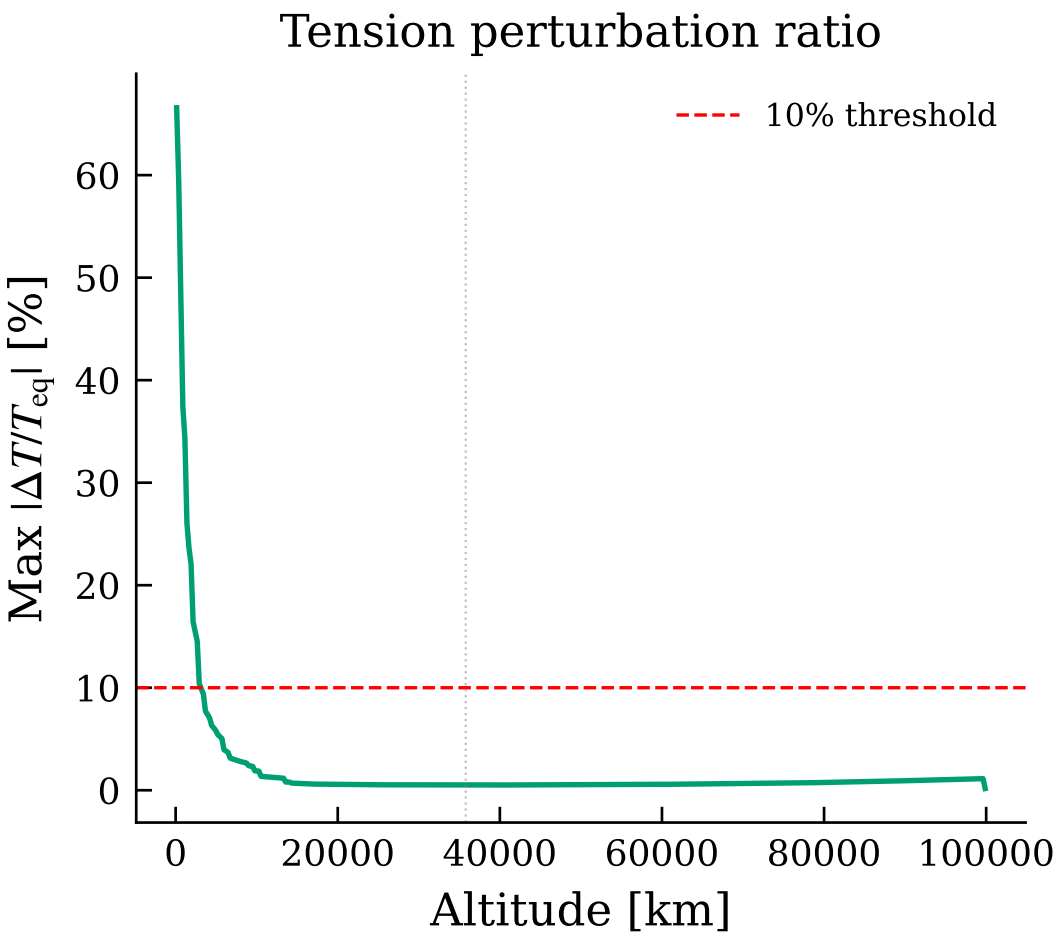

Figure 19: Tension perturbation ratio $\Delta T/T_{\mathrm{eq}}$ vs. altitude.

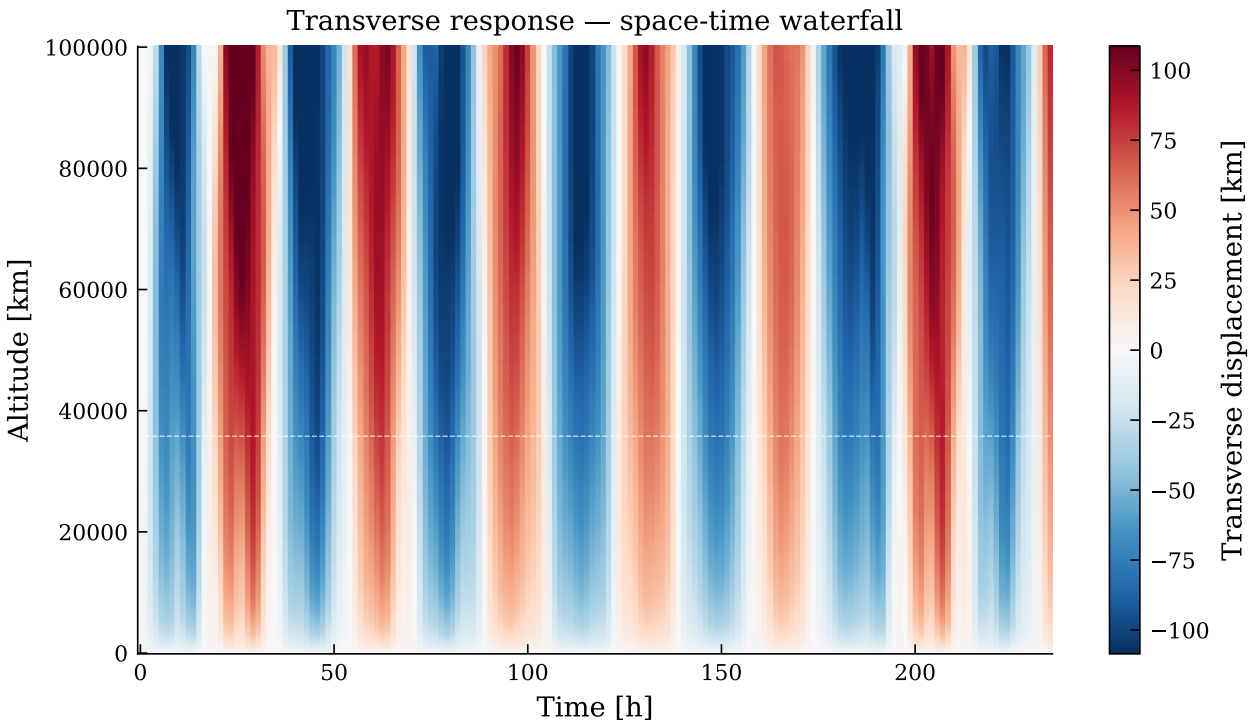

Figure 20: Transverse response space-time waterfall $v(r, t)$.

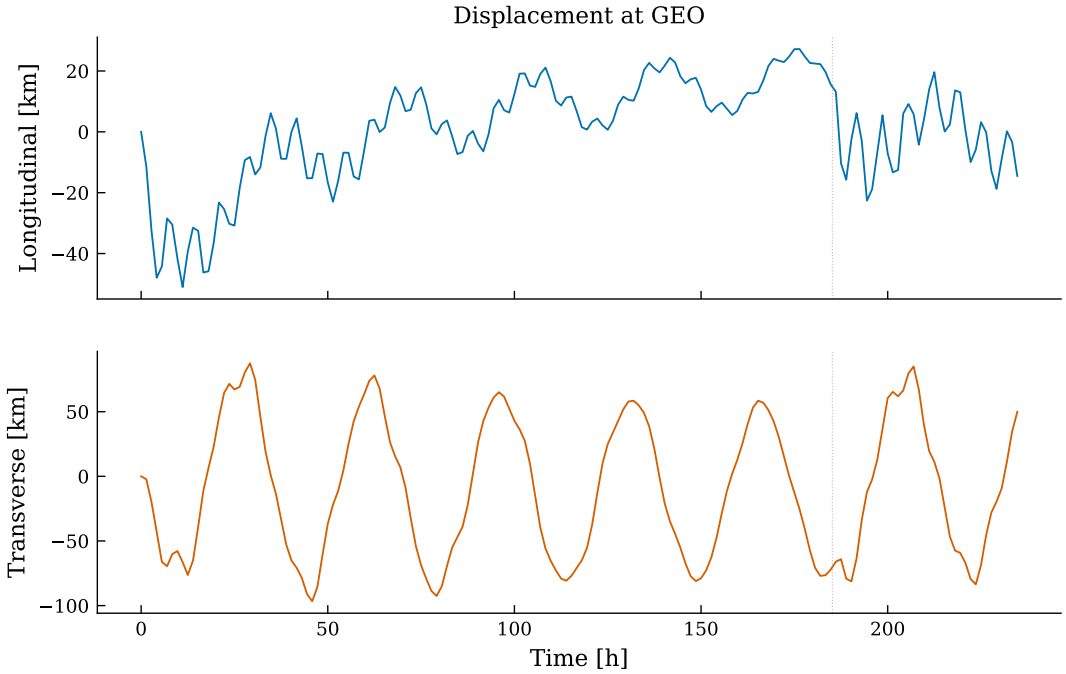

Figure 21: Displacement time history at GEO (u and v).

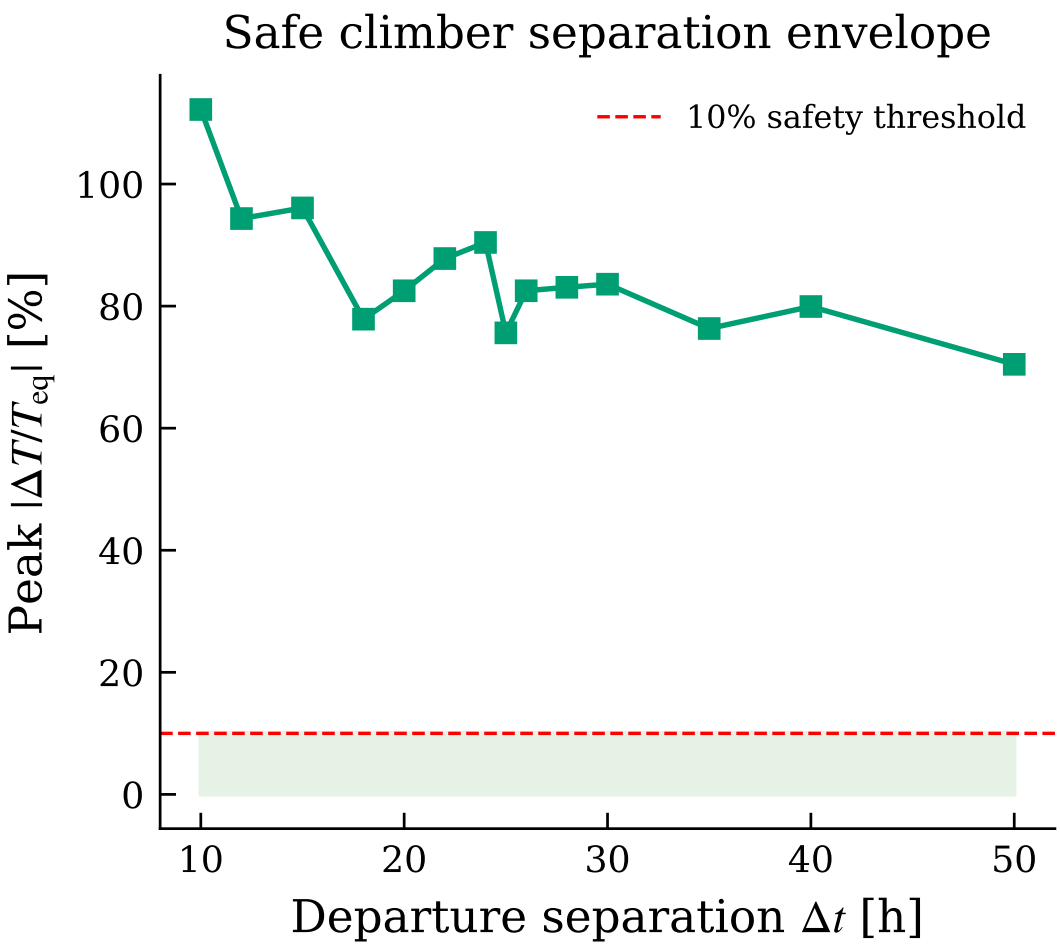

Figure 22: Tension perturbation ratio vs. departure separation.

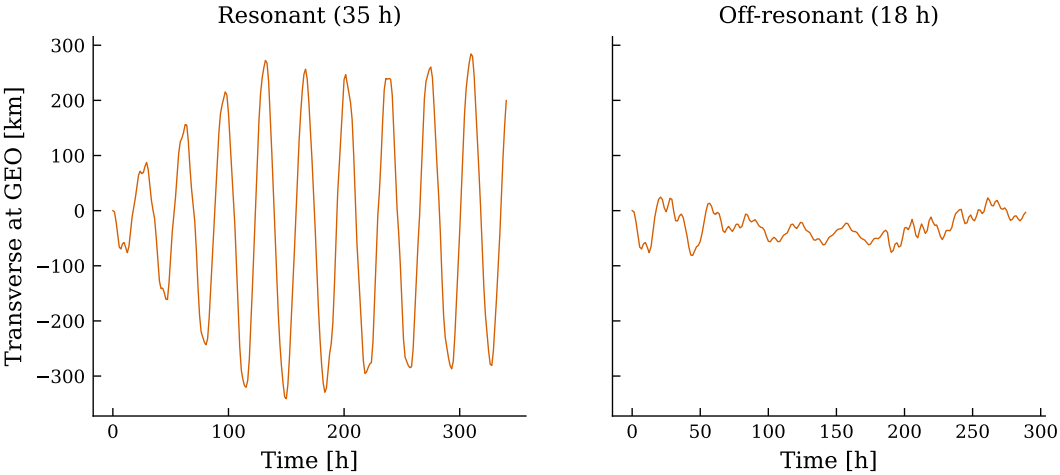

Figure 23: GEO transverse displacement: resonant vs. off-resonant.

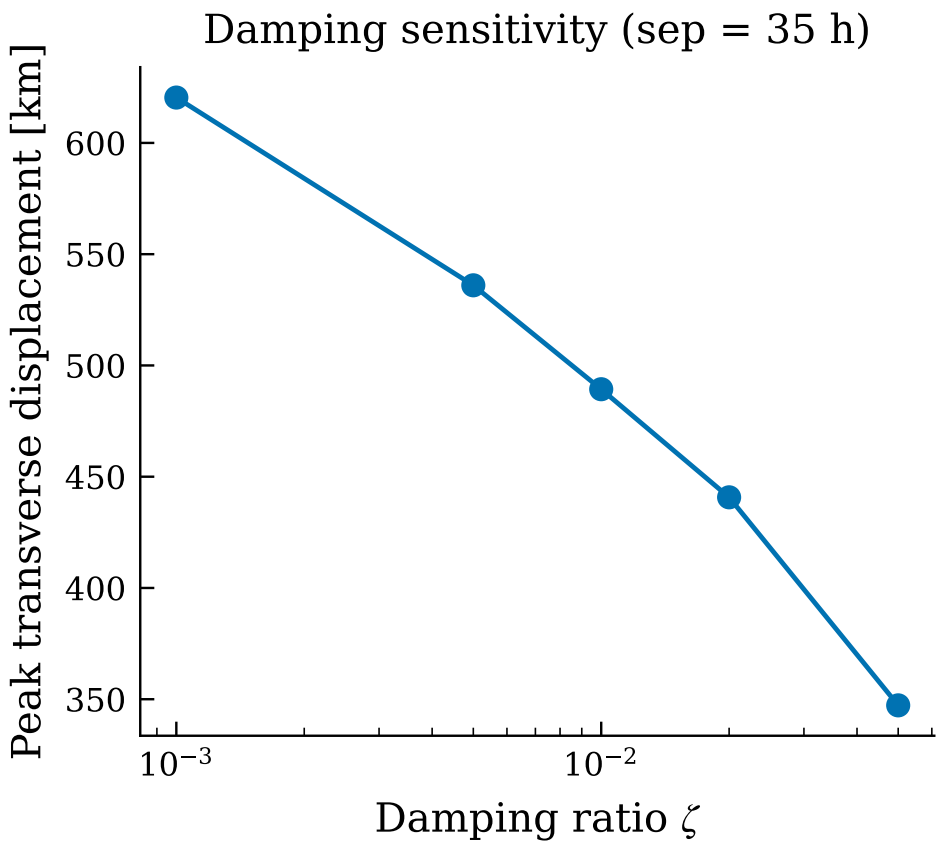

Figure 24: Peak transverse displacement vs. damping ratio ζ.

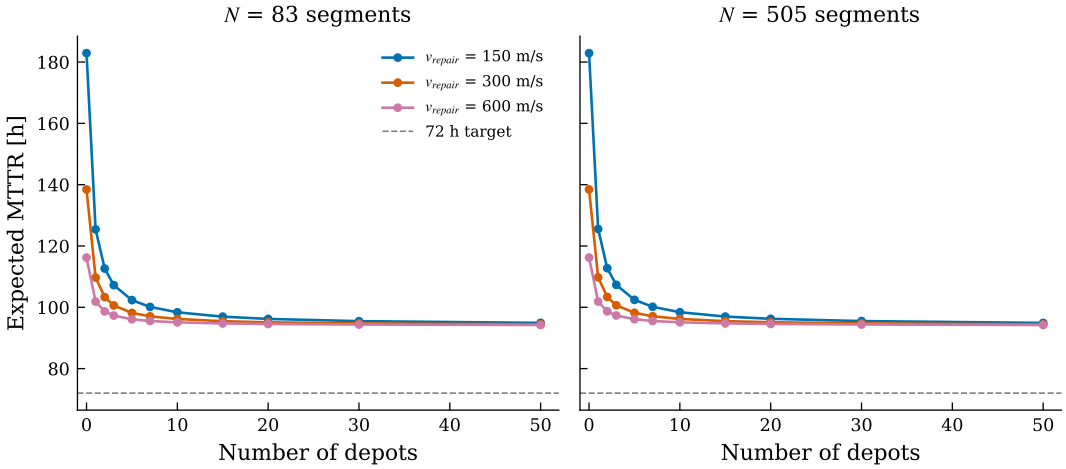

Figure 25: Expected MTTR vs. depot count at $v_{\text{repair}} = \{150, 300, 600\}$ m/s.

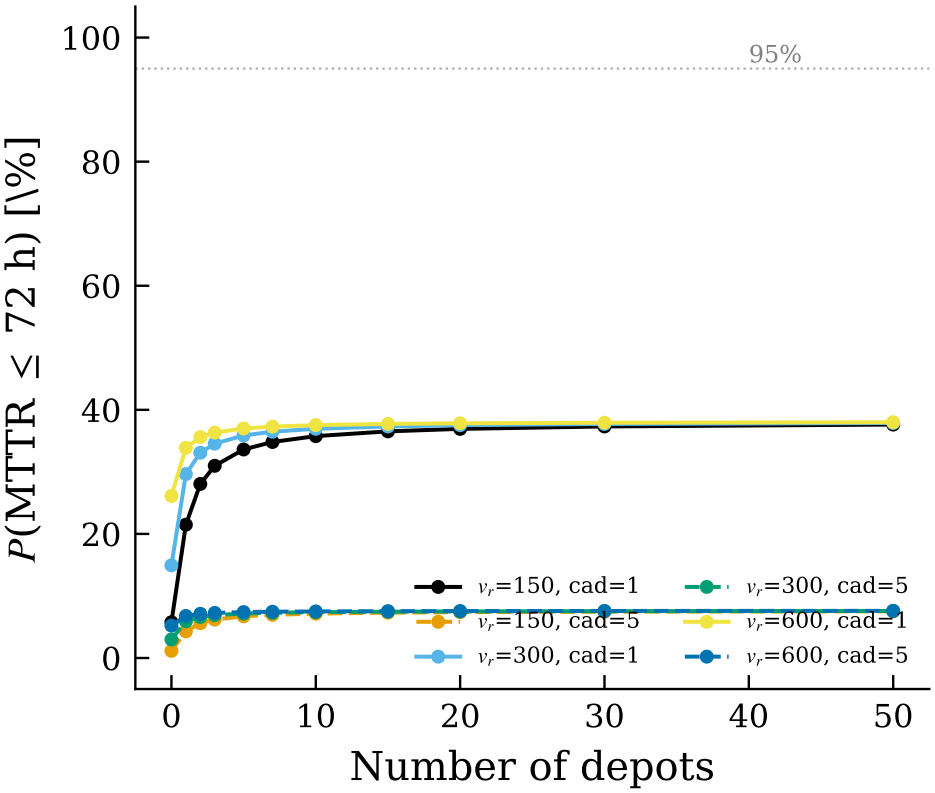

Figure 26: $P(\text{MTTR} \leq 72\text{ h})$ vs. depot count.

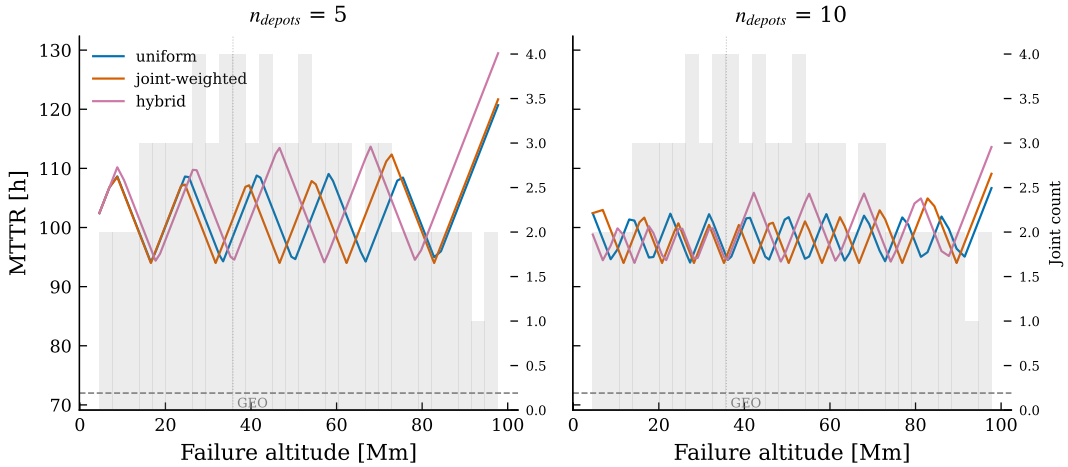

Figure 27: MTTR vs. failure altitude for three placement strategies.

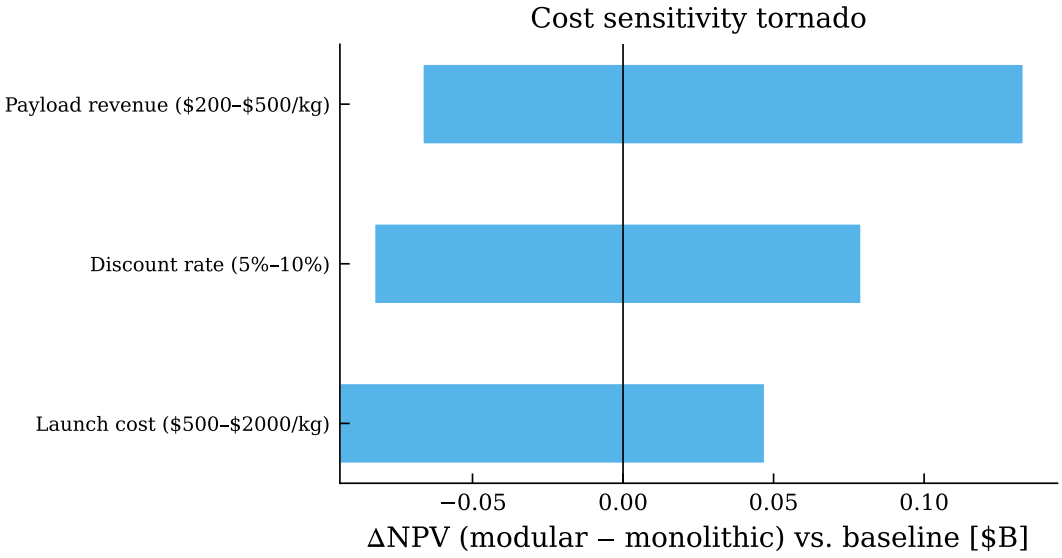

Figure 28: Cost sensitivity tornado diagram.

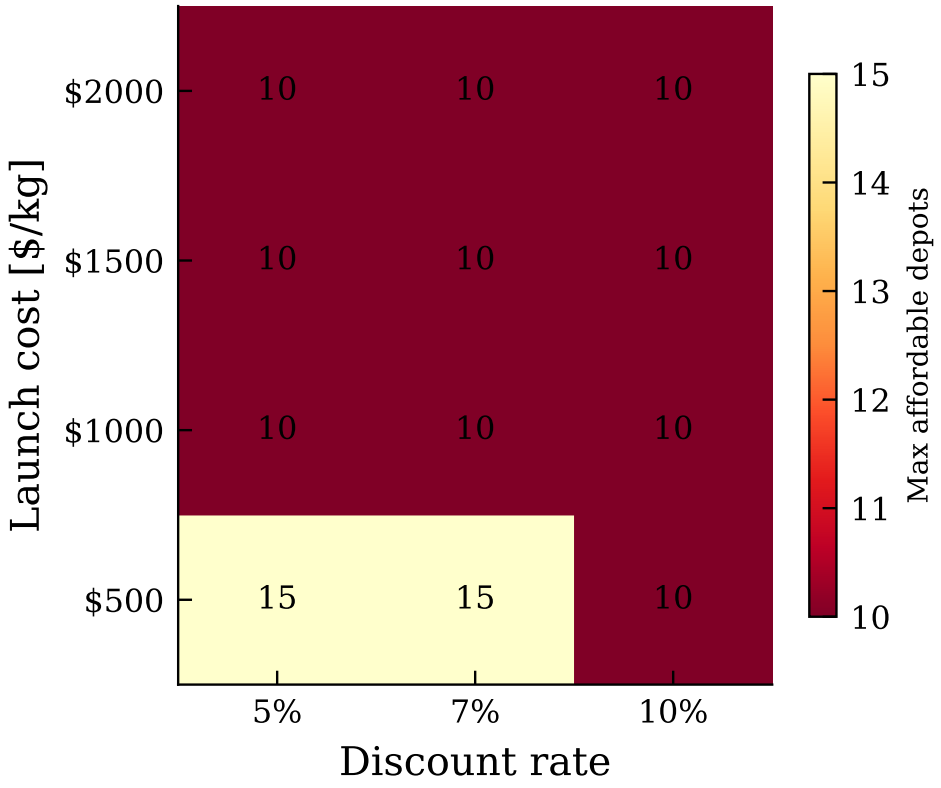

Figure 29: Maximum affordable depots vs. scenario parameters.

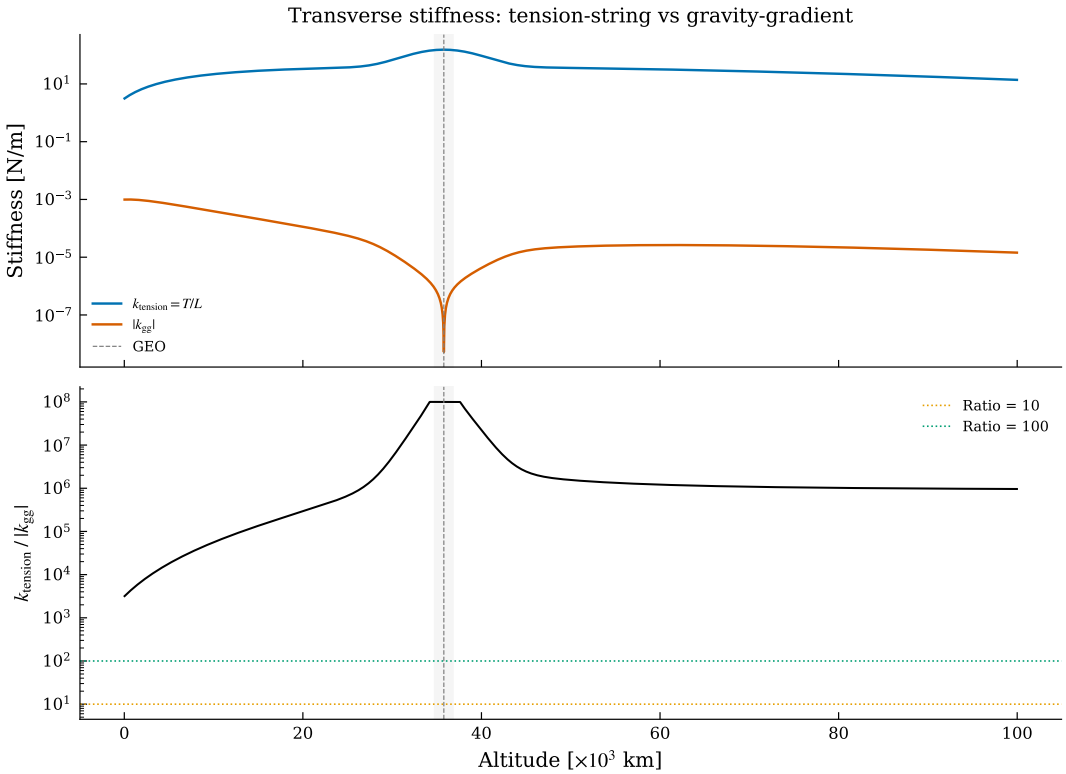

Figure 30: Tension vs. gravity-gradient stiffness comparison (2-panel).

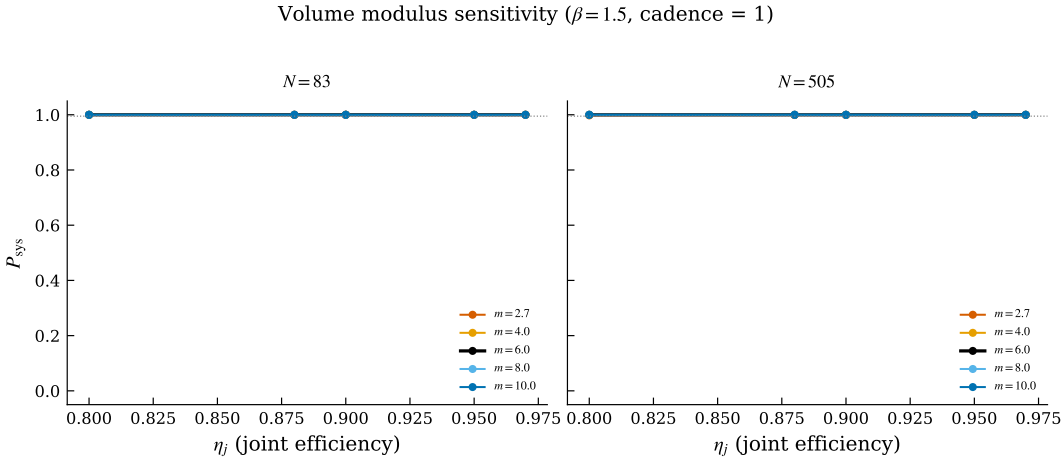

Figure 31: P_{sys} vs. η_j at Weibull volume moduli $m \in \{2.7, 4, 6, 8, 10\}$.